TP 9 : Apache kafka

Université / École :

- Université Ibn Tofail, Kénitra

Encadrement pédagogique :

- Pr. Hajar Filali

Rédigé par :

- Mohammed Jebrou
- Hajar Farid

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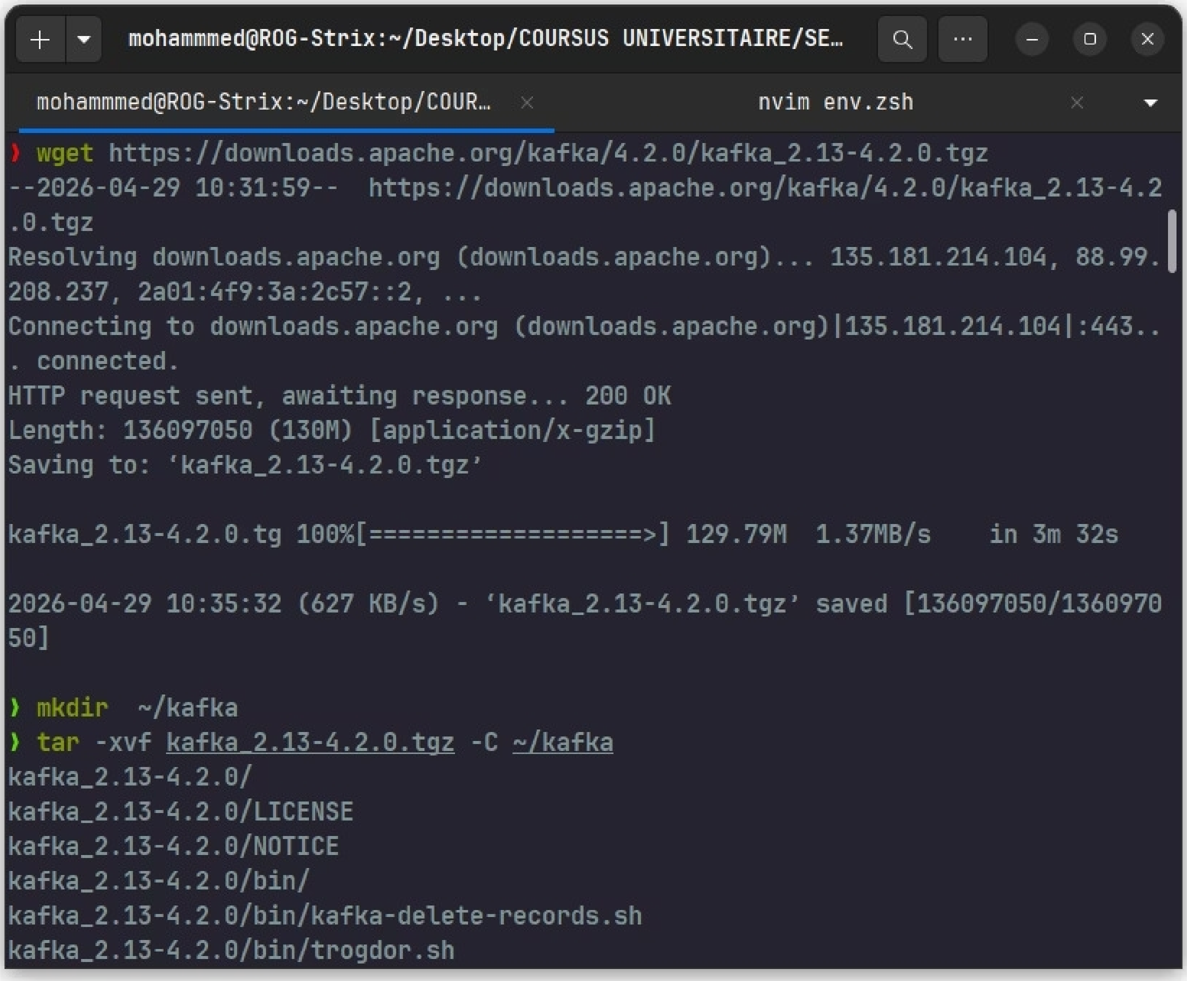
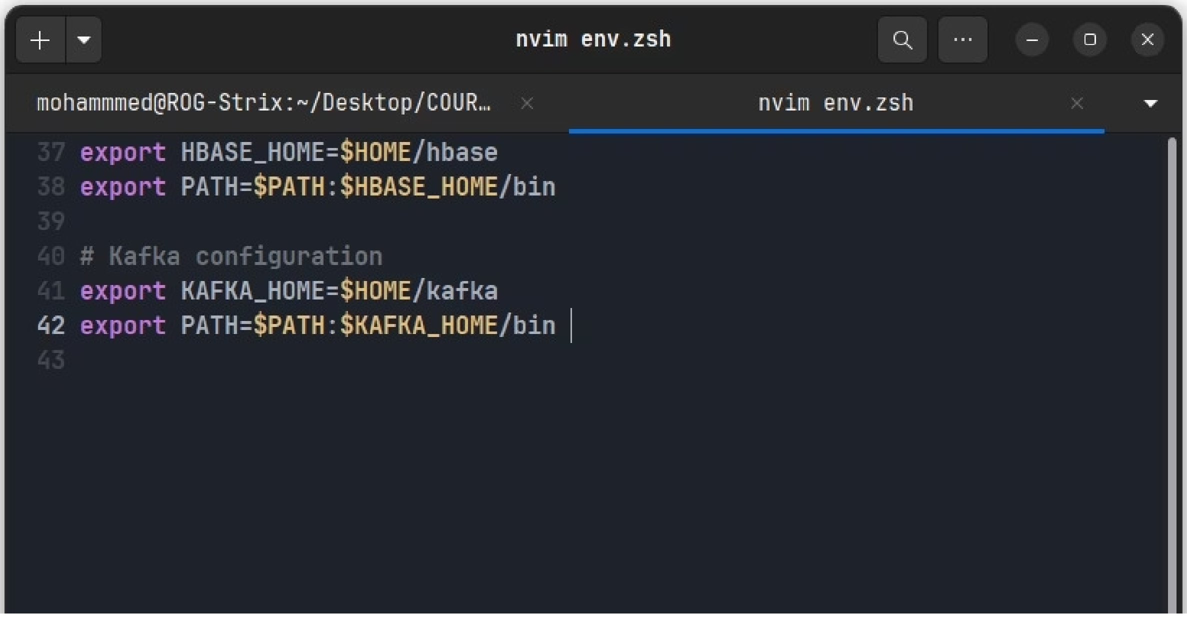
Installtion du Kafka

1. Téléchargement et extraction des fichiers



```
mohammmed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE…
mohammmed@ROG-Strix:~/Desktop/COUR…    nvim env.zsh
❯ wget https://downloads.apache.org/kafka/4.2.0/kafka_2.13-4.2.0.tgz
--2026-04-29 10:31:59--  https://downloads.apache.org/kafka/4.2.0/kafka_2.13-4.2.0.tgz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 136097050 (130M) [application/x-gzip]
Saving to: 'kafka_2.13-4.2.0.tgz'

kafka_2.13-4.2.0.tg 100%[===================>] 129.79M  1.37MB/s    in 3m 32s

2026-04-29 10:35:32 (627 KB/s) - 'kafka_2.13-4.2.0.tgz' saved [136097050/136097050]

❯ mkdir  ~/kafka
❯ tar -xvf kafka_2.13-4.2.0.tgz -C ~/kafka
kafka_2.13-4.2.0/
kafka_2.13-4.2.0/LICENSE
kafka_2.13-4.2.0/NOTICE
kafka_2.13-4.2.0/bin/
kafka_2.13-4.2.0/bin/kafka-delete-records.sh
kafka_2.13-4.2.0/bin/trogdor.sh
```

2. Configuration des variable d'envirenement



```
nvim env.zsh
mohammmed@ROG-Strix:~/Desktop/COUR…    nvim env.zsh
export HBASE_HOME=$HOME/hbase
export PATH=$PATH:$HBASE_HOME/bin

# Kafka configuration
export KAFKA_HOME=$HOME/kafka
export PATH=$PATH:$KAFKA_HOME/bin
```

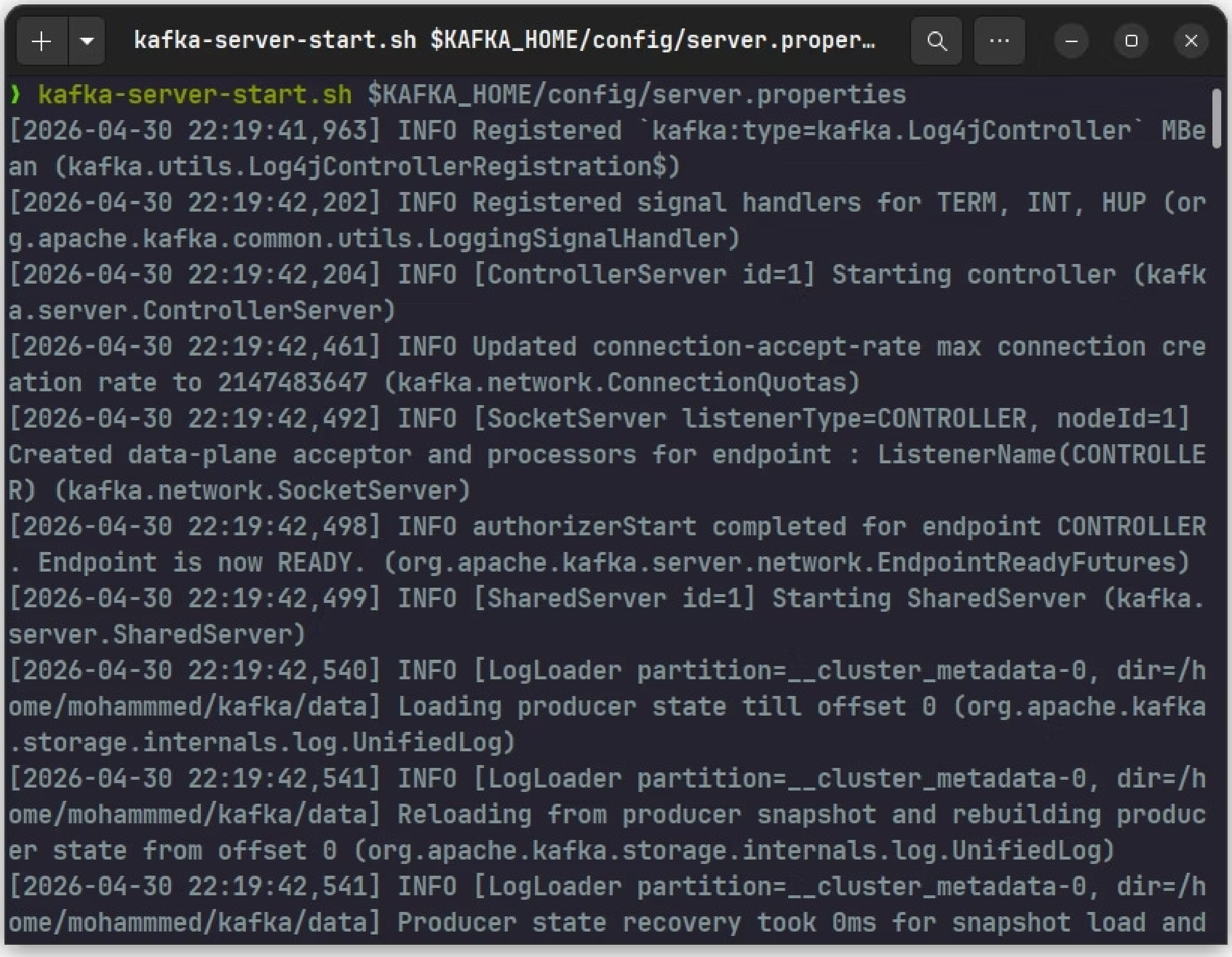
3. Configuration du fichier de serveur config/server.properties

```
socket.send.buffer.bytes=102400

# The receive buffer (SO_RCVBUF) used by the socket server
socket.receive.buffer.bytes=102400

# The maximum size of a request that the socket server will accept (protection against OOM)
socket.request.max.bytes=104857600


############################# Log Basics #############################

# A comma separated list of directories under which to store log files
log.dirs=/home/mohammmed/kafka/data
# The default number of log partitions per topic. More partitions allow greater
# parallelism for consumption, but this will also result in more files across
# the brokers.
num.partitions=1

# The number of threads per data directory to be used for log recovery @@@
```

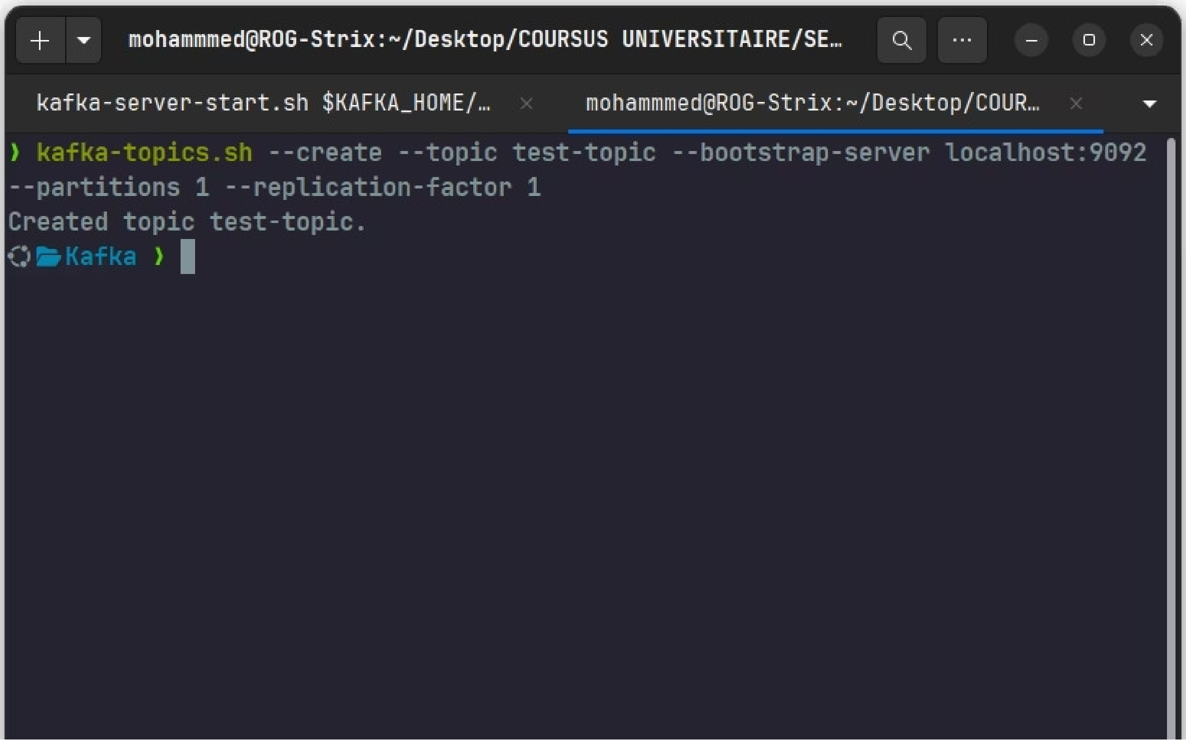
4. installation du cluster   et formatage du stockage a travers le id généré

```
❯ KAFKA_CLUSTER_ID="$(kafka-storage.sh random-uuid)"
❯ kafka-storage.sh format --standalone -t $KAFKA_CLUSTER_ID -c $KAFKA_HOME/config/server.properties
Bootstrap metadata: BootstrapMetadata(records=[ApiMessageAndVersion(FeatureLevelRecord(name='metadata.version', featureLevel=29) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='eligible.leader.replicas.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='group.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='share.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='streams.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='transaction.version', featureLevel=2) at version 0)], metadataVersionLevel=29, source=format command)
Formatting dynamic metadata voter directory /home/mohammmed/kafka/data with metadata.version 4.2-IV1.
Kafka ❯
```

5. Démarche du Broker Kafka

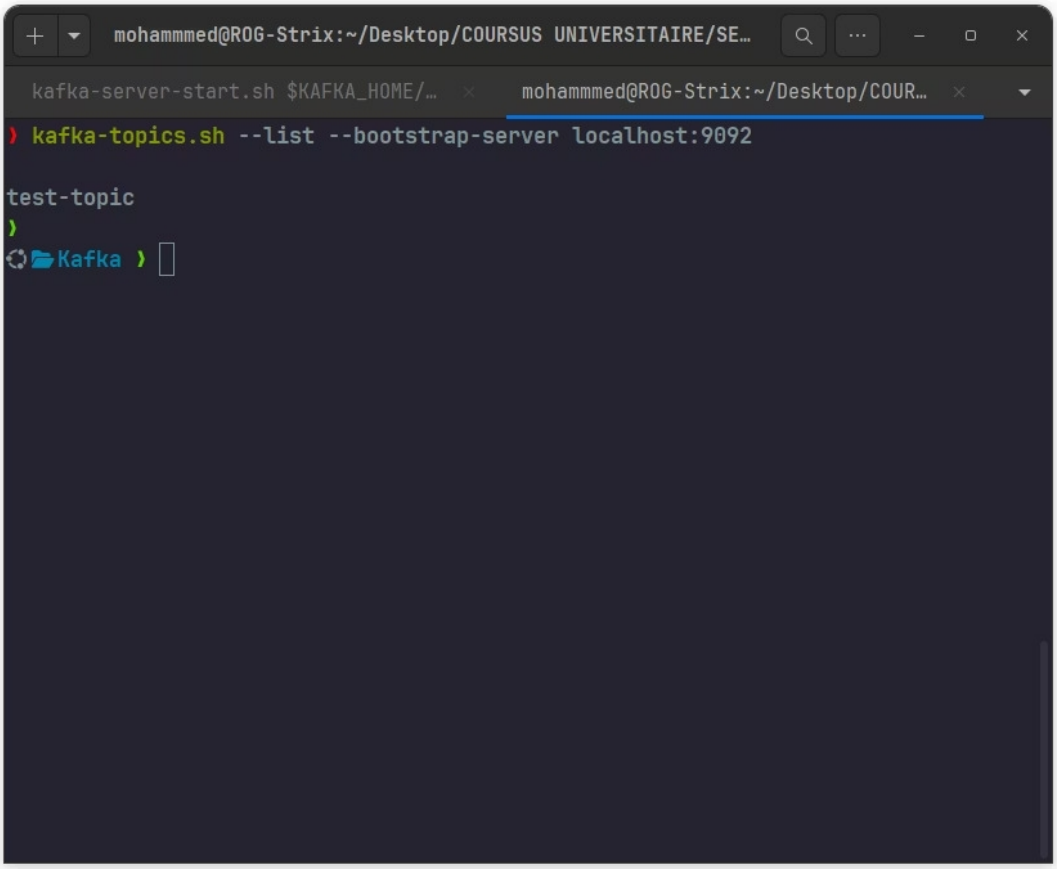


6. Création d'un topic

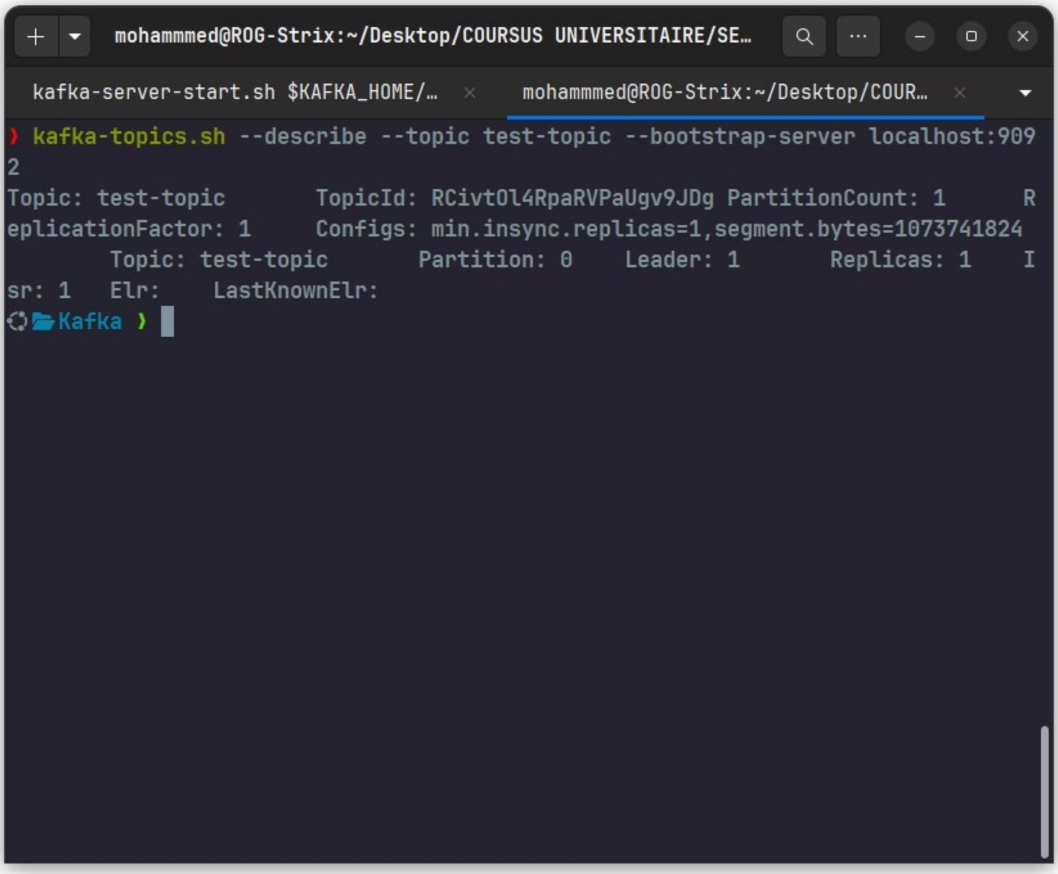


7. Verifier les topics

a. Lister les topics



b.décrire topics



8. Installation du du Kafka pythn

```
mohammmed@ROG-Strix:~/Desktop/COURSUS UNIVERSITAIRE/SE…
kafka-server-start.sh $KAFKA_HOME/…    mohammmed@ROG-Strix:~/Desktop/COUR…
❯ pip install kafka-python confluent-kafka
Collecting kafka-python
  Downloading kafka_python-2.3.1-py2.py3-none-any.whl.metadata (9.5 kB)
Collecting confluent-kafka
  Downloading confluent_kafka-2.14.0-cp312-cp312-manylinux_2_28_x86_64.whl.metad
ata (32 kB)
Downloading kafka_python-2.3.1-py2.py3-none-any.whl (326 kB)
Downloading confluent_kafka-2.14.0-cp312-cp312-manylinux_2_28_x86_64.whl (4.0 MB
)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.0/4.0 MB 813.6 kB/s  0:00:04
Installing collected packages: kafka-python, confluent-kafka
Successfully installed confluent-kafka-2.14.0 kafka-python-2.3.1
Kafka ❯                                           took ⌛ 8s kafka
```

9. Crétion du producer pyrhon

```
nvim producer.py
kafka-server-start.sh $KAFKA_HOME/…    nvim producer.py
from kafka import kafkaProducer
import time

producer = kafkaProducer(bootsrap_servers ='localhost:9092')

for i in range( 10):
    message= f'Hello From Producer {i+1}'
    producer.send('test-topic',value= message.encode())
    print(f'Envoyé:{message}')
    time.sleep(1)
producer.flush()

INSERT  producer.py                                          11/17
```

cusomer python

```python
from kafka import KafkaConsumer

consumer= KafkaConsumer('test-topic', bootstrap_servers= 'localhost:9092',
                        auto_offset_reset='earliest')

for maessage in consumer:
    print(maessage.value.decode())
```

Résultat finale

```
❯ python producer.py
Envoyé:Hello From Producer 1
Envoyé:Hello From Producer 2
Envoyé:Hello From Producer 3
Envoyé:Hello From Producer 4
Envoyé:Hello From Producer 5
Envoyé:Hello From Producer 6
Envoyé:Hello From Producer 7
Envoyé:Hello From Producer 8
Envoyé:Hello From Producer 9
Envoyé:Hello From Producer 10
Kafka ❯                                   took ⌛ 10s kafka
```

```
❯ python consumer.py
Hello From Producer 1
Hello From Producer 2
Hello From Producer 3
Hello From Producer 4
Hello From Producer 5
Hello From Producer 6
Hello From Producer 7
Hello From Producer 8
Hello From Producer 9
Hello From Producer 10
```